Project Initialization and Planning Phase

Date	21 October 2026
Team ID	Piyush-Balodiya-GenAI-2026
Project Name	Multimodal Construction Intelligence Platform (CONSTRUCTAI)
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love. A well-articulated customer problem statement allows you and your team to find the ideal solution for your customers' challenges. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A Site Safety Officer & EHS Auditor on an active site.	Enforce zero OSHA/IS non-compliance and monitor PPE usage (helmets, vests, harnesses).	Manual site rounds miss intermittent safety violations in high-risk zones.	CCTV feeds and on-site workforce across multi-acre sites make continuous human monitoring impossible.	Anxious about preventable site injuries, worker accidents, and statutory penalties.
PS-2	A Structural Engineer & Quality Control Inspector.	Accurately assess concrete surface cracks and categorize structural integrity risks against IS 456 limits.	Manual crack gauge measurements are slow, error-prone, and lack instant remediation plans.	Visual checks cannot immediately calculate shear vs shrinkage severity or estimate repair costs on-site.	Vulnerable to missing critical structural damage and delayed retrofitting interventions.
PS-3	A Project Planning Engineer & Construction Manager.	Generate accurate material quantity takeoffs (BOQ), milestones, and weather risk contingencies.	Manual BOQ estimation takes days and unexpected monsoon rains halt concrete pours without warning.	Site logs, IS code guidelines, and weather alerts are fragmented across disjointed tools.	Overwhelmed by budget overruns, resource waste, and avoidable schedule delays.